

■ FM (Frequency Modulation) – Complete Revision Notes

◆ 1. Basics of Angle Modulation

◆ General Form

$$x(t) = A \cos(\omega_c t + \phi(t))$$

- $\phi(t) \rightarrow$ phase term carrying information

◆ Types

✓ FM (Frequency Modulation)

$$\phi(t) = K_f \int m(t) dt$$

$$f_i(t) = f_c + K_f m(t)$$

👉 Frequency varies with message

✓ PM (Phase Modulation)

$$\phi(t) = K_p m(t)$$

$$f_i(t) = f_c + \frac{K_p}{2\pi} \frac{dm(t)}{dt}$$

👉 Frequency depends on slope of message

🔥 Relation Between FM & PM

$$\text{FM}(m(t)) = \text{PM}\left(\int m(t) dt\right)$$

$$\text{PM}(m(t)) = \text{FM}\left(\frac{dm(t)}{dt}\right)$$

◆ 2. Single Tone FM

Assume:

$$m(t) = A_m \cos(\omega_m t)$$

◆ Standard FM Equation

$$x(t) = A \cos(\omega_c t + \beta \sin \omega_m t)$$

◆ Modulation Index

$$\beta = \frac{\Delta f}{f_m} = \frac{K_f A_m}{\omega_m}$$

◆ Frequency Deviation

$$\Delta f = \text{max deviation of frequency}$$

◆ 3. Instantaneous Frequency

$$f_i(t) = \frac{1}{2\pi} \frac{d\theta(t)}{dt}$$

For FM:

$$f_i(t) = f_c + K_f m(t)$$

For PM:

$$f_i(t) = f_c + \frac{K_p}{2\pi} \frac{dm(t)}{dt}$$

◆ 4. FM Spectrum

◆ Expansion

$$x(t) = A \sum_{m=-\infty}^{\infty} J_m(\beta) \cos((\omega_c + m\omega_m)t)$$

◆ Key Points

- Infinite sidebands
- Frequencies:

$$f_c \pm m f_m$$

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- Amplitude:

$$J_m(\beta)$$

◆ Bessel Properties

$$J_{-m}(\beta) = (-1)^m J_m(\beta)$$

◆ Power

$$P = \frac{A^2}{2}$$

$$\sum J_m^2(\beta) = 1$$

- 👉 Power redistributed, not increased

◆ 5. Bandwidth of FM

◆ Problem

- Infinite spectrum → impractical

◆ Carson's Rule

$$B = 2(\Delta f + f_m)$$

◆ Alternate Form

$$B = 2(\beta + 1)f_m$$

◆ Interpretation

- $\Delta f \rightarrow$ spread due to modulation
- $f_m \rightarrow$ spacing of sidebands

◆ 6. NBFM vs WBFM

◆ Narrowband FM (NBFM)

Condition:

$$\beta \ll 1$$

Approximation:

$$J_0 \approx 1, \quad J_1 \approx \frac{\beta}{2}, \quad J_{m \geq 2} \approx 0$$

Signal:

$$x(t) \approx A \cos \omega_c t + \text{two sidebands}$$

👉 Similar to AM

Bandwidth:

$$B \approx 2f_m$$

◆ Wideband FM (WBFM)

Condition:

$$\beta \gg 1$$

Features:

- Many sidebands
- Large bandwidth
- Better noise immunity

◆ 7. FM Generation

◆ (A) Direct Method (VCO)

Concept:

$$f \propto \frac{1}{\sqrt{LG}}$$

Using Varactor Diode:

$$C = C_0 - Km(t)$$

Result:

$$\omega_i(t) = \omega_0 + K_f m(t)$$

Advantages:

- ✓ Simple
- ✓ Direct FM

Disadvantages:

- ✗ Frequency drift
- ✗ Not stable

◆ (B) Indirect Method (Armstrong)

Steps:

1. Generate NBFM
2. Use frequency multipliers
3. Use mixer to shift frequency

Key Concept:

$$\cos(\omega_c t + \phi(t)) \rightarrow \cos(n\omega_c t + n\phi(t))$$

So:

$$\beta_{new} = n\beta$$

Advantages:

- ✓ Very stable
- ✓ Used in broadcasting

◆ 8. Noise in FM

◆ SNR

$$\text{SNR} = \frac{\text{Signal power}}{\text{Noise power}}$$

◆ FM vs AM

Feature	FM	AM
Noise immunity	High	Low
Bandwidth	Large	Small

◆ Reason

- 👉 Noise affects amplitude
- 👉 FM depends on frequency

◆ 9. FM Demodulation

◆ Goal

Recover:

$$m(t)$$

◆ (1) Differentiator Method

$$\frac{dx}{dt} = -A(\omega_c + K_f m(t)) \sin(\dots)$$

Idea:

FM → differentiate → becomes AM → detect

Problem:

✗ Amplifies noise

◆ (2) Limiter + BPF + Detector

Blocks:

1. Limiter → remove amplitude noise
2. BPF → remove harmonics
3. Differentiator
4. Envelope detector

Output:

$$m(t)$$

◆ (3) Slope Detector

Idea:

- Use sloped frequency response
- Convert frequency → amplitude

Improvement:

- 👉 Staggered tuned circuits

$$V_o \propto y_1 - y_2$$

◆ (4) PLL Demodulator (Most Important)

Blocks:

- Phase detector
- Loop filter
- VCO

Working:

$$e(t) \propto \sin(\phi - \theta)$$

In lock condition:

$$\phi \approx \theta$$

Output:

$$e(t) \propto \frac{d\phi}{dt} \propto m(t)$$

Advantages:

- ✓ Accurate
- ✓ Noise resistant
- ✓ Widely used

◆ 10. PLL Model

◆ Nonlinear:

$$e(t) \propto \sin(\phi - \theta)$$

◆ Linear Approximation:

$$\sin x \approx x$$

◆ Transfer Function:

$$\frac{\theta(s)}{\phi(s)} = \frac{AKH(s)}{s + AKH(s)}$$

◆ Lock Condition:

$$\theta \approx \phi$$

🔥 FINAL QUICK RECALL (1 MIN REVISION)

- 👉 FM = frequency variation
 - 👉 $\beta = \Delta f / f_m$
 - 👉 Infinite sidebands → use Carson
 - 👉 NBFM $\approx$ AM
 - 👉 WBFM → many sidebands
 - 👉 VCO → simple but unstable
 - 👉 Armstrong → stable
 - 👉 Limiter removes amplitude noise
 - 👉 Differentiator → FM → AM
 - 👉 PLL → best demodulator
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